Notation. Let

$$P(x) = x^7 + a_6x^6 + a_5x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0$$

with $a_i \in \mathbb{Z}$. By hypothesis all coefficients are odd, and

$$|a_i| < 2025^2 \qquad (i = 0, 1, \ldots, 7).$$

Moreover, for every i we have $P(|a_i|) = 0$.

0.0.1 1. First consequences of the condition

Since $|a_i|$ is an integer root of P, the Rational Root Theorem tells us that $|a_i|$ divides the constant term a_0. Hence

$$|a_i| \mid a_0 \qquad (i = 0, 1, \ldots, 7).$$

Define

$$M = \max\{|a_i| : 0 \le i \le 7\}.$$

Because M itself is one of the numbers $|a_i|$, (1) gives $M \mid a_0$. Write $a_0 = M \cdot k$ with $k \in \mathbb{Z}$. Then $|a_0| = |k|\, M$. But $|a_0| \le M$ (by definition of M), so $|k| \le 1$. Since $a_0 \ne 0$ (it is odd), $k \ne 0$; therefore $|k| = 1$ and

$$a_0 = \varepsilon M, \qquad \varepsilon = \pm 1.$$

Thus $M = |a_0|$ and M is odd (because a_0 is odd).

The hypothesis applied to $|a_7| = 1$ yields $P(1) = 0$; hence 1 is a root. Applied to $|a_0| = M$ we obtain $P(M) = 0$.

0.0.2 2. The case $M = 1$

If $M = 1$, then every $|a_i| \le 1$ and, being odd positive integers, they must all equal 1. Hence $a_i \in \{\pm 1\}$ for $i = 0, \ldots, 6$.

The condition $P(1) = 0$ becomes

$$1 + a_6 + a_5 + a_4 + a_3 + a_2 + a_1 + a_0 = 0 \iff a_6 + a_5 + a_4 + a_3 + a_2 + a_1 + a_0 = -1.$$

Thus we need seven numbers each ± 1 whose sum is -1. This happens exactly when four of them are -1 and three are $+1$. The number of such tuples $(a_0, \ldots, a_6)$ is $\binom{7}{3} = 35$.

All these polynomials clearly satisfy the required conditions (all coefficients are odd, $|a_i| = 1 < 2025^2$, and 1 is a root). Hence there are 35 **good polynomials with** $M = 1$.

0.0.3 3. The case $M > 1$

Now M is an odd integer > 1; therefore $M \geq 3$.

Because M is a root we can factor

$$P(x) = (x - M)\, Q(x),$$

where Q is a monic polynomial of degree 6 with integer coefficients. Then

$$Q(0) = \frac{P(0)}{-M} = \frac{a_0}{-M} = -\varepsilon = \pm 1.$$

Suppose some $|a_i|$ satisfies $d > 1$ and $d \neq M$. Then d is a root of P and $d \neq M$; consequently $(x - d)$ must divide $Q(x)$, so d is an integer root of Q. By the Rational Root Theorem applied to Q, d divides $Q(0) = \pm 1$, which forces $d = \pm 1$, contradicting $d > 1$. Therefore **every** $|a_i|$ is either 1 or M. Hence

$$a_i \in \{\pm 1, \pm M\} \qquad (i = 0, 1, \ldots, 6).$$

We already know $a_0 = \varepsilon M$ with $\varepsilon = \pm 1$. The two equations $P(1) = 0$ and $P(M) = 0$ give

$$\sum_{i=0}^{6} a_i = -1,$$
$$M^7 + a_6 M^6 + a_5 M^5 + a_4 M^4 + a_3 M^3 + a_2 M^2 + a_1 M + a_0 = 0.$$

Substituting $a_0 = \varepsilon M$ into (6) and dividing by M ($M \neq 0$) we obtain

$$\varepsilon + a_1 + a_2 M + a_3 M^2 + a_4 M^3 + a_5 M^4 + a_6 M^5 = -M^6.$$

3.1 Solving the system We analyse (7) modulo M. All terms containing M are multiples of M; hence

$$\varepsilon + a_1 \equiv 0 \pmod{M}.$$

Because $\varepsilon = \pm 1$ and $a_1 \in \{\pm 1, \pm M\}$, we examine the possibilities. If $a_1 = \pm M$ then $\varepsilon + a_1 = \pm 1 \pm M$; its absolute value is $|M \pm 1|$, which is not a multiple of M (since $M \geq 3$ and $M \nmid (M \pm 1)$). Thus a_1 cannot be $\pm M$. Therefore $a_1 = \pm 1$, and (8) forces $\varepsilon + a_1 = 0$; consequently

$$a_1 = -\varepsilon.$$

Insert $a_1 = -\varepsilon$ into (7):

$$\begin{aligned} &\varepsilon + (-\varepsilon) + a_2 M + a_3 M^2 + a_4 M^3 + a_5 M^4 + a_6 M^5 = -M^6 \\ \implies \quad & a_2 M + a_3 M^2 + a_4 M^3 + a_5 M^4 + a_6 M^5 = -M^6. \end{aligned}$$

Divide by M:

$$a_2 + a_3M + a_4M^2 + a_5M^3 + a_6M^4 = -M^5.$$

Modulo M we get

$$a_2 \equiv 0 \pmod{M}.$$

Since $a_2 \in \{\pm 1, \pm M\}$, (11) forces $a_2 = \pm M$. Write $a_2 = \delta_2 M$ with $\delta_2 = \pm 1$.

Substituting into (10) and dividing by M:

$$\delta_2 + a_3 + a_4M + a_5M^2 + a_6M^3 = -M^4.$$

Modulo M gives $\delta_2 + a_3 \equiv 0 \pmod{M}$. As before, $a_3 \in \{\pm 1, \pm M\}$; if $a_3 = \pm M$ then $\delta_2 + a_3$ is not a multiple of M. Hence $a_3 = \pm 1$ and we must have $\delta_2 + a_3 = 0$; thus

$$a_3 = -\delta_2.$$

Now (12) becomes

$$\delta_2 + (-\delta_2) + a_4M + a_5M^2 + a_6M^3 = -M^4$$
$$\implies \quad a_4M + a_5M^2 + a_6M^3 = -M^4.$$

Divide by M:

$$a_4 + a_5M + a_6M^2 = -M^3.$$

Modulo M yields $a_4 \equiv 0 \pmod{M}$; therefore $a_4 = \pm M$. Set $a_4 = \delta_4 M$ with $\delta_4 = \pm 1$.

Insert into (14) and divide by M:

$$\delta_4 + a_5 + a_6M = -M^2.$$

Modulo M gives $\delta_4 + a_5 \equiv 0 \pmod{M}$. Consequently $a_5 = \pm 1$ and $\delta_4 + a_5 = 0$; i.e.

$$a_5 = -\delta_4.$$

Now (15) simplifies to

$$\delta_4 + (-\delta_4) + a_6M = -M^2 \implies a_6M = -M^2 \implies a_6 = -M.$$

Thus we have expressed all coefficients in terms of three signs:

$$
\begin{aligned}
a_0 &= \varepsilon M, & \varepsilon &= \pm 1,\\
a_1 &= -\varepsilon, & &\\
a_2 &= \delta_2 M, & \delta_2 &= \pm 1,\\
a_3 &= -\delta_2, & &\\
a_4 &= \delta_4 M, & \delta_4 &= \pm 1,\\
a_5 &= -\delta_4, & &\\
a_6 &= -M. & &
\end{aligned}
$$

3.2 Using the sum condition (5)

$$
\begin{aligned}
\sum_{i=0}^{6} a_i &= \varepsilon M - \varepsilon + \delta_2 M - \delta_2 + \delta_4 M - \delta_4 - M\\
&= M(\varepsilon + \delta_2 + \delta_4 - 1) - (\varepsilon + \delta_2 + \delta_4).
\end{aligned}
$$

Set $S = \varepsilon + \delta_2 + \delta_4$. Equation (5) becomes

$$
M(S-1) - S = -1 \implies MS - M - S = -1 \implies (M-1)S - M = -1.
$$

Since $M > 1$, $M - 1 \neq 0$; dividing gives

$$
(M-1)S = M - 1 \implies S = 1.
$$

Thus

$$
\varepsilon + \delta_2 + \delta_4 = 1.
$$

Each of $\varepsilon, \delta_2, \delta_4$ is ± 1; their sum can be $-3, -1, 1, 3$. The condition $S = 1$ is satisfied exactly when two of them are $+1$ and one is -1. There are $\binom{3}{2} = 3$ such triples $(\varepsilon, \delta_2, \delta_4)$.

Consequently, for **each** odd $M > 1$ there are exactly **3** choices of signs, hence exactly **3** distinct polynomials satisfying all requirements.

3.3 Counting the admissible M The coefficients must satisfy $|a_i| < 2025^2$. Because all $|a_i|$ are either 1 or M, this condition is equivalent to $M < 2025^2$. Moreover, M is odd (as noted after (2)). Hence we need to count the odd integers M with

$$
1 < M < 2025^2.
$$

Compute

$$
2025^2 = 4\,100\,625.
$$

The odd numbers strictly between 1 and $4\,100\,625$ are

$$3,\ 5,\ 7,\ \ldots,\ 4\,100\,623.$$

The number of such integers is

$$\frac{4\,100\,625-3}{2}=\frac{4\,100\,622}{2}=2\,050\,311.$$

(Indeed, the count of odd integers from 1 to 4 100 623 inclusive is $(4\,100\,623-1)/2+1=2\,050\,312$; removing $M=1$ leaves 2 050 311.)

For each of these 2 050 311 values of M we obtain 3 polynomials, giving

$$3\times 2\,050\,311=6\,150\,933$$

good polynomials with $M>1$.

0.0.4 4. Total number of good polynomials

Adding the 35 polynomials from the case $M=1$ we obtain

$$\boxed{6\,150\,968}.$$
